

Computational Physics

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Lecture #21

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Nonlinear Dynamics

- nonlinear (systems of) equations important in:

physics, math, engineering, climate, ...

- difficult to solve

- generate interesting phenomena:

chaotic behavior, multi-stability, complex patterns in spacetime

two goals:

(1) discuss methods for numerical solutions

- Newton's method (Newton-Raphson)

(2) analysis of equations analytically, to gain physical insights into allowed solutions and their behavior

- important when we no longer have analytic solutions e.g. 2-body,

- or even, no conserved quantities e.g. E_{tot} in 3-body Schrödinger cases

↳ e.g. when [macroscopic] dissipation/heat transfer occurs, converting into microscopic DOFs, e.g. turbulent energy cascade

e.g. energy loss via radiation in astrophysical plasmas

problem types:

recurrence relations: $x_{n+1} = f(x_n)$

e.g. Logistic map, either (i) dynamics is truly discrete, or (ii) in any numerical solution even to a continuous process

ODEs: $y'(t) = f(t, y)$

e.g. Newton's second law of motion

systems of ODEs: $\begin{cases} x'(t) = f(x, y) \\ y'(t) = g(x, y) \end{cases}$

e.g. population dynamics, where x, y represent two different species

PDEs: $\frac{\partial^2 y}{\partial t^2} = f(y) \frac{\partial^2 y}{\partial x^2}$ so $y = y(x, t)$

e.g. heat, wave, Euler/Navier-Stokes
↳ $f = c^2$

Def: a function $f(x)$ is linear if:

$f(x+y) = f(x) + f(y)$ "additivity" or "superposition principle"

$f(ax) = a f(x)$ "homogeneity"

otherwise, $f(x)$ is nonlinear

Def: a number p is a fixed point ("stationary point")

(2)

$$\text{of } f(x) \text{ if } \boxed{f(p) = p}$$

" p is invariant under f , since f maps p to itself"

note: if p exists, can define a new function $g(x) = x - f(x)$ such that $g(p) = 0$

↳ fixed points and roots are equivalent characteristics of an equation

↳ understanding $f(x)$ near p is similar to understanding $g(x)$ near zero, i.e. differential equations near their solutions

Def: a number p is a fixed point of a differential equation $y'(t) = f(t, y)$

$$\text{if } y'(t=p) = \left. \frac{dy}{dt} \right|_p = \boxed{y'(p) = 0}$$

Def: any system of nonlinear equations has the form

$$\frac{dx_1}{dt} = f_1(x_1, x_2, \dots, x_n) = 0$$

$$\frac{dx_2}{dt} = f_2(x_1, x_2, \dots, x_n) = 0$$

$$\vdots$$
$$\frac{dx_n}{dt} = f_n(x_1, x_2, \dots, x_n) = 0$$

$$\Rightarrow \boxed{\vec{F}(\vec{x}) = 0}$$

$$\text{where } \vec{F}: \mathbb{R}^n \rightarrow \mathbb{R}^n \\ = (f_1, f_2, \dots, f_n)$$

Def: $\vec{F}$ has a fixed point $\vec{p} \in \mathbb{R}^n$ if $\boxed{\vec{F}(\vec{p}) = \vec{p}}$

Population Dynamics

③

- simple ODEs demonstrate nonlinear behavior and analysis techniques

Ex: Malthus (1798)

- model interactions between species, and population growth

note: similar ODEs for radioactive decay or chemical networks, but here they are nonlinear!

$$\frac{dN}{dt} = (b-d)N$$
$$= rN$$

where

$N \equiv$ number of population members
[e.g. humans on Earth, kangaroos in Australia, some closed ecosystem]

$b, d \equiv$ birth, death rates, per unit time

so $\frac{dN}{dt}$ is the net growth rate:

- $b > d$ ($r > 0$) $\rightarrow$ growth
- $d > b$ ($r < 0$) $\rightarrow$ decline

for convenience, make dimensionless by defining:

$$\tau = rt \quad n = N/N_0 \quad \text{where } N_0 = N(t_0=0)$$

$$\Rightarrow \boxed{\frac{dn}{d\tau} = n(\tau)} \quad \text{has the well-known solution } n(\tau) = Ce^{\tau}$$

[at $t=0$, $\tau=0$, $n=1 \Rightarrow C=1$ in our dimensionless case]

$\hookrightarrow$ exponential, and unlimited, growth so long as $r > 0$

Ex: Verhulst (1836)

- idea: finite resources limit growth

$$\frac{dN}{dt} = rN(1 - N/K)$$

limiting factor, where $K \equiv$ limiting value for N

three regimes:

(1) $N < K$: growth, and $N'(t) > 0$

(2) $N = K$: constant, and $N'(t) = 0$

(3) $N > K$: decline, and $N'(t) < 0$

again a dimensionless form for convenience: $\tau = rt$, $n = N/K$

$$\Rightarrow \boxed{\frac{dn}{d\tau} = n(1-n)} \quad \text{is nonlinear in } n, \text{ unlike Malthus}$$

Q: What are the fixed points of these ODEs?
(n^*)

$\frac{dn}{dt} = 0$ i.e. solution is fixed / stationary / constant, does not evolve

- tempting to say that n^* is a "stable" value of the population, but this is not necessarily true
- rather, mathematically, if $n = n^*$ exactly, then n will not change

Malthus: $[n^* = 0]$ is a "trivial" FP

Verhulst: $\begin{bmatrix} n^* = 0 \\ n^* = 1 \end{bmatrix}$ is a "trivial" FP
is when $N = K$

Q: What happens near a FP? Clear that $n^* = 0$ FPs are unstable $\rightarrow$ divergent,

Stability

while $n^* = 1$ FP is stable, i.e. if the solution is slightly perturbed, it returns to it

idea: use Taylor expansion to study small deviation from the FP:

for $\frac{dn}{dt} = f(n)$ consider $n = n^* + \delta_n$ where $\delta_n / n \ll 1$

$$\begin{aligned} \text{RHS: } f(n) &= f(n^* + \delta_n) \\ &= \cancel{f(n^*)} + \delta_n f'(n^*) + \cancel{O(\delta_n^2)} \quad \text{drop} \\ &\quad \text{zero by definition} \\ &= \delta_n f'(n^*) \end{aligned}$$

$$\begin{aligned} \text{LHS: } \frac{dn}{dt} &= \frac{d}{dt}(n^* + \delta_n) \\ &= \cancel{\frac{dn^*}{dt}} + \frac{d\delta_n}{dt} = \frac{d}{dt}\delta_n \end{aligned}$$

$\Rightarrow \frac{d}{dt}\delta_n = \delta_n f'(n^*)$ is a simple ODE with an analytical solution

$$\Rightarrow \delta_n(t) = \delta_n(0) \exp[f'(n^*)t]$$

the sign of $\text{Re}[f'(n^*)]$ determines if the δ_n perturbation $\uparrow$ or $\downarrow$ i.e. is (unstable) or (stable)